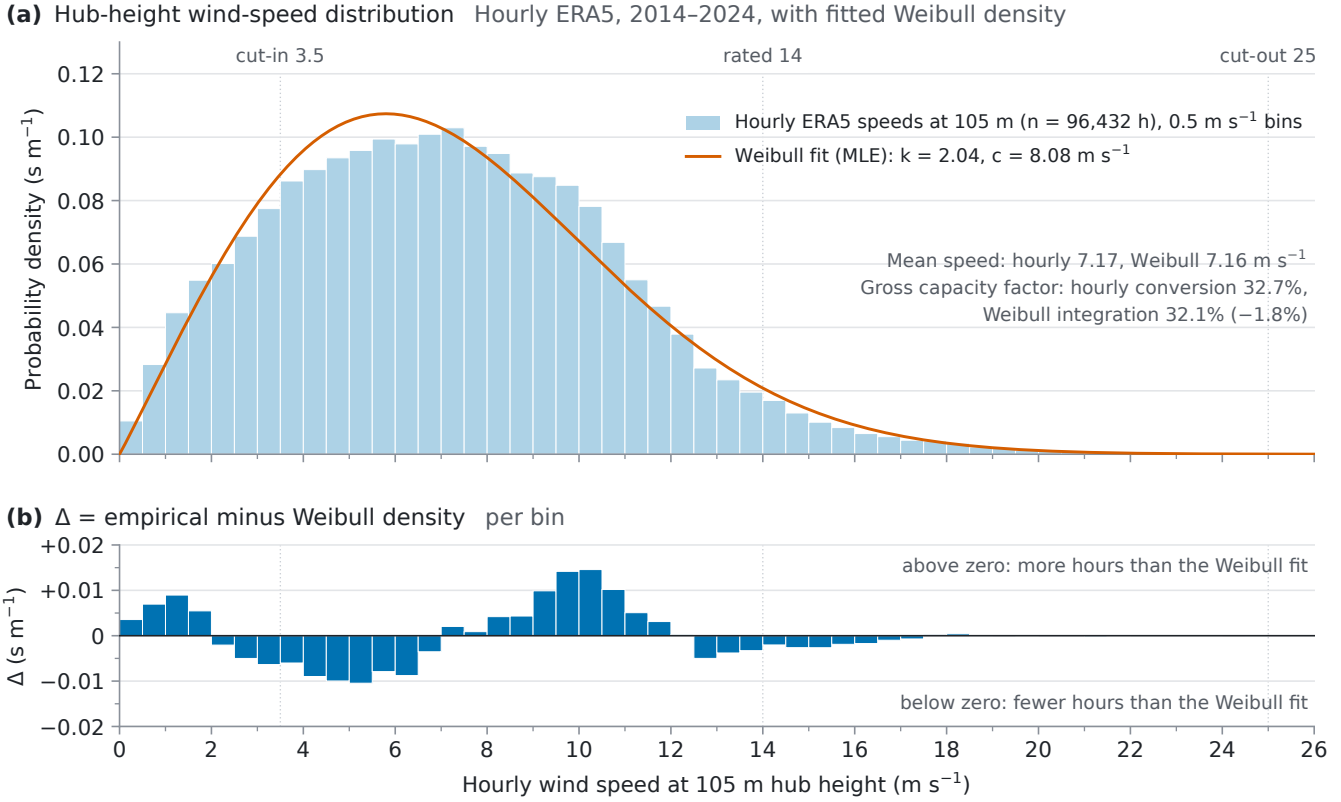
(a) Hub-height wind-speed distribution
Hourly ERA5, 2014–2024, with fitted Weibull density
cut-in 3.5
rated 14
cut-out 25
Probability density (s m⁻¹)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Hourly ERA5 speeds at 105 m (n = 96,432 h), 0.5 m s⁻¹ bins
Weibull fit (MLE): k = 2.04, c = 8.08 m s⁻¹
Mean speed: hourly 7.17, Weibull 7.16 m s⁻¹
Gross capacity factor: hourly conversion 32.7%,
Weibull integration 32.1% (−1.8%)
(b) Δ = empirical minus Weibull density
per bin
Δ (s m⁻¹)
+0.02
+0.01
0
−0.01
−0.02
above zero: more hours than the Weibull fit
below zero: fewer hours than the Weibull fit
0
2
4
6
8
10
12
14
16
18
20
22
24
26
Hourly wind speed at 105 m hub height (m s⁻¹)